Conduct and Testing For Conduct

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Fall 2026

Grad IO

The problem

Conduct Testing in Industrial Organization

Foundational Empirical IO Question: How do we observe data on price and quantity and infer which model of firm behavior generated those outcomes?

- Early work: Porter (1983), Bresnahan (1982,1987)
- Subsequent work defined the "menu" approach: Nevo (1998, 2001), Villas-Boas (2007)
- Recent revival of "internalization" parameters: Miller and Weinberg (2017), Crawford, Lee, Whinston, and Yurukoglu (2017), Pakes (2017)
- Parallel work by: Duarte, Magnolfi, Sølvsten, Sullivan (QE 2024): which test is best (RV) and when instruments are too weak to test. Magnolfi, Quint, Sullivan, Waldfogel (2022): should we test or estimate?
- Applications of our test: Starc and Wollmann (2022), Scuderi and Roussille (2025), others?
- Today: the classic approaches, then the Backus–Conlon–Sinkinson scalar test ("BCS 1.0") and the full conditional-moment test ("BCS 2.0"), then the common-ownership application: what identifies conduct, and what the tests find.

Is conduct testable? Berry and Haile (2014): yes.

Conduct Testing in Industrial Organization

- Absent additional restrictions, we cannot generally look at data on (P, Q) and decide whether or not collusion is taking place.
  - You say we started colluding at date t, I say we received a correlated shock to mc.
- We can make progress in two ways: (1) parametric restrictions on marginal costs; (2) exclusion restrictions on supply.
  - Most of the literature focuses on (1) by assuming something like: $\ln mc_{jt} = \mathsf{x}_{jt}\gamma_1 + \mathsf{w}_{jt}\gamma_2 + \omega_{jt}$.
  - In principle (2) is possible if we have instruments that shift demand for products but not supply. (These are much easier to come up with than "supply shifters"). (This is the point Berry Haile 2014 make).

A famous plot (Bresnahan 87)

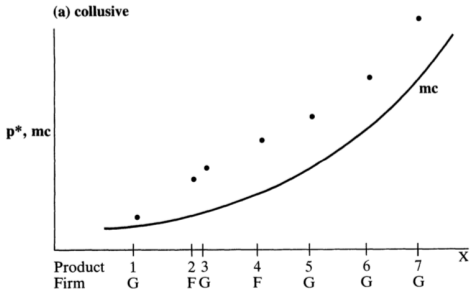


Figure 2(a)

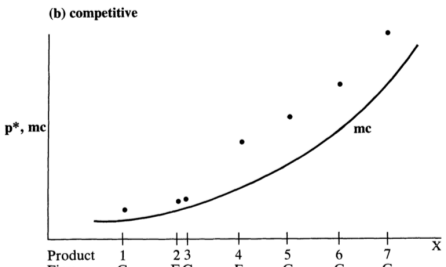


Figure 2(b)

Bresnahan (1980/1982) recognized this problem: we need “rotations of demand”.

Conduct Testing in Pictures (Berry Haile 2014)

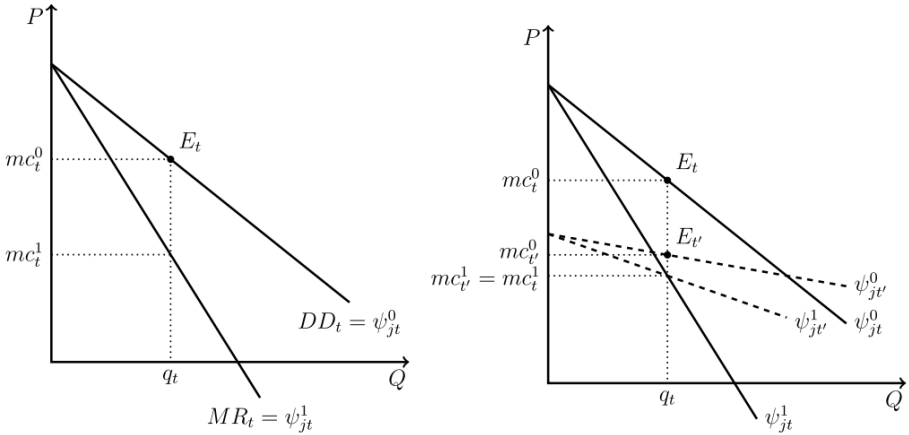


Figure 2(ab) from Berry and Haile (2014), Example 1.

Setup: Notation and Utility

We begin with a relatively standard BLP-style differentiated products setup.

- Markets t
- Products j
- Data $\chi_t = \{(\mathsf{x}_{jt}, \mathsf{v}_{jt}, \mathsf{w}_{jt}) \text{ for all } j \in \mathcal{J}_t\}$.
- Market Shares $\mathbf{s}_t = [s_{1t}, \dots, s_{Jt}, s_{0t}]$.
- Prices $\mathbf{p}_t = [p_{1t}, \dots, p_{Jt}]$.
- Consumers i with demographics y_{it} (income, presence of kids, etc.)

Testing Conduct: Multiproduct Bertrand Example

We generalize the $\mathcal{H}(\kappa)$ and derive multi-product Bertrand FOCs:

$$\arg\max_{p \in \mathcal{J}_f} \pi_f(\mathbf{p}) = \sum_{j \in \mathcal{J}_f} (p_j - c_j) \cdot q_j(\mathbf{p}) + \kappa_{fg} \sum_{j \in \mathcal{J}_g} (p_j - c_j) \cdot q_j(\mathbf{p})$$

$$\rightarrow 0 = q_j(\mathbf{p}) + \sum_{k \in (\mathcal{J}_f, \mathcal{J}_g)} \kappa_{fg} \cdot (p_k - c_k) \frac{\partial q_k}{\partial p_j}(\mathbf{p})$$

- Instead of 0's and 1's we now have $\kappa_{fg} \in [0, 1]$ representing how much firm f cares about the profits of g.
  - If f and g merge (or fully coordinated) then $\kappa_{fg} = 1$
  - Often in the real world firms cannot reach fully collusive profits and $\kappa_{fg} \in (0, 1)$.
  - Evidence that $\kappa_{fg} > 0$ is not necessarily evidence of malfeasance, just a deviation from static Bertrand pricing (e.g. Cournot, Dynamics, etc.)

Testing Conduct: Multiproduct Bertrand Example

- Recall the Δ matrix which we can write as $\Delta = \tilde{\Delta} \odot \mathcal{H}(\kappa)$, where $\odot$ is the element-wise or Hadamard product of two matrices.
  - $\tilde{\Delta}$ is the matrix of demand derivatives with $\Delta(j,k) = \frac{\partial q_j}{\partial p_k}$ for all elements.
  - $\mathcal{H}(\kappa) = \kappa_{fg}$ for products owned by (f,g) where $\kappa_{ff} = 1$ always.
- Mergers are about changing 0's to 1's in the $\mathcal{H}(\kappa)$ matrix.
- Matrix form of FOC: $q(\mathbf{p}) = \Delta(\mathbf{p}, \kappa) \cdot (\mathbf{p} - \mathbf{mc})$
- $\mathbf{mc} = \mathbf{p} - \underbrace{\Delta(\mathbf{p}, \theta_2, \kappa)^{-1}\boldsymbol{\sigma}(\mathbf{p})}_{\eta(\mathbf{p},\mathbf{s},\theta_2,\kappa)}$ where η_{jt} is the markup.

What is the Common Ownership Hypothesis?

- Economics starts from the premise that firms maximize profits.
  - Friedman (1953): natural selection of firms and billiards players.
  - Or, firms answer to investors, maximize shareholder value.
  - We generally think of these objectives as aligned (possibly subject to managers' agency frictions).
- So what do investors want?
  - Some (diversified) investors may hold stakes in you *and your competitor*. These are called "common owners."
  - Common owners are harmed when managers engage in business stealing.
- As a theory of firm behavior in joint ventures this is an old idea. The recent innovation is to extend this approach to passive or institutional investors

Some Reasonable (?) Assumptions (BCS 2020 P&P)

Most models of common ownership (implicitly) make the following assumptions:

Investor Portfolios Investors are entitled to a share of profits β_{fs} and hold portfolios $V_s = \sum_f \beta_{fs}\pi_f$

Manager Agency (Rotemberg 1984) Managers maximize a (γ) weighted average of investor payoffs $Q_f = \sum_s \gamma_{fs} V_s$.

These are pretty general since s could contain the manager herself.

Derivation of Profit Weight

Easy to show:

$$
\begin{aligned}
Q_f\left(x_f, x_{-f}\right) &= \sum_{\forall s} \gamma_{fs} \cdot V_s\left(x_f, x_{-f}\right) \\
&\propto \pi_f + \sum_{g \neq f} \underbrace{\left(\frac{\sum_{\forall s} \gamma_{fs}\beta_{gs}}{\sum_{\forall s} \gamma_{fs}\beta_{fs}}\right)}_{\equiv \kappa_{fg}\left(\gamma_f, \beta\right)} \pi_g
\end{aligned}
$$

- κ_{fg} is interpreted as a profit weight, where one dollar of profits at firm g are valued as κ_{fg} dollars in the objective function of firm f.
- Depends on two primitives: β and γ.
  - β (cash-flow rights) is often directly observable.
  - γ embeds a model of corporate governance. Literature assumes $\gamma = \beta$ ("proportional control").
- Not specific to a particular strategic game: this is the $\mathcal{H}(\kappa)$ from our multiproduct Bertrand FOCs.

Reasons for Deviations from Multiproduct Bertrand

Biased estimates of own and cross price derivatives: For anything to work, you have correct estimates of $\tilde{\Delta}$. My prior is most papers underestimate diversion ratios for close substitutes.

Vertical Relationships: Who sets supermarket prices? Just the retailer? Just the manufacturer? Some combination of both? Retailers tend to soften downstream price competition.

Faulty Timing Assumptions: Bertrand is a simultaneous move pricing game. Lots of alternatives (Stackelberg leader-follower, Edgeworth cycles, etc.).

Dynamics and Dynamic Pricing: Forward looking firms or consumers might not set static Nash prices. [e.g. Temporary Sales, Switching Costs, Network Effects, etc.]

Unmodeled Supergame: Maybe firms are legally tacitly colluding, higher prices might be about what firms believe will happen in a price war.

Simultaneous Problem

Assume additivity, and write in terms of structural errors:

$$\delta_{jt}(\mathbf{s}_t, \mathsf{y}_t, \tilde{\theta}_2) + \alpha p_{jt} = h_d(\mathsf{x}_{jt}, \mathsf{v}_{jt}, \theta_1) + \xi_{jt}$$
$$f\left(p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa)\right) = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}, \theta_3) + \omega_{jt}$$

- To simplify slides we let $f(x) = x$ (often $f(x) = \log(x)$) but we can put that in $h_s(\cdot)$.
- $h(\cdot)$ are often just linear relationships like: $\theta_1[\mathsf{x}_{jt}, \mathsf{v}_{jt}]$.
- Endogeneity Problem: p_{jt} and η_{jt} are functions of $(\boldsymbol{\xi}, \boldsymbol{\omega})$.
- (θ_2, κ) parameters that determine markups

Two Ways In (Recap of the BLP Lecture)

1. **Demand first.** Estimate θ_2 from demand alone with $\mathbb{E}[\xi_{jt} \mid \mathsf{x}_t, \mathsf{v}_t, \mathsf{w}_t] = 0$; recover $\widehat{\mathbf{mc}} = \mathbf{p} - \boldsymbol{\eta}(\mathbf{p}, \mathbf{s}, \widehat{\theta}_2, \kappa)$ for any κ.
2. **Jointly.** Stack the supply moments $\mathbb{E}[\omega_{jt} \mid \mathsf{x}_t, \mathsf{w}_t, \mathsf{v}_t] = 0$ with demand; the supply side sharpens α (BLP 1995) but imports whatever conduct you assumed.

The problem with either: given $[\mathbf{q}, \mathbf{p}, \widetilde{\Delta}, \mathcal{H}(\kappa)]$ there is always a vector $\mathbf{mc}$ that rationalizes the data (J equations, J unknowns). Nonparametrically κ is not identified without more.

- Restriction 1: a model of marginal cost, $f(p_{jt} - \eta_{jt}(\kappa)) = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}; \theta_3) + \omega_{jt}$, so that "$mc < 0$" or "$mc$ jumps around" become evidence.
- Restriction 2: exclusion. Something that moves markups but not costs, $\mathbb{E}[\omega_{jt} \mid z_{jt}] = 0$ (Berry Haile 2014: demand rotators v_t, rivals' characteristics).
- Everything that follows is a way of turning those two restrictions into a test. The second one does all the real work.

Classic approaches

Estimation vs. Menus

There are 2.5 ways to think about conduct:

1. Using moment conditions to estimate $\widehat{\kappa}$ or $\mathcal{H}(\kappa)$ directly. (Wald)
    - Often with a small number of parameters (ie: $\kappa_{fg} = 0$ except for firms I know are in a cartel).
    - Can be challenging to tell similar values of κ_{fg} apart (under-powered).
2. "Menu Approach" / (LR)
    - Nevo (Economics Letters 1998)
    - Bresnahan (1987)
    - Compare some goodness-of-fit critieria across assumed values of κ (Bertrand vs. Collusion)
3. Rejecting (or failing to reject) a single model.
    - "We can reject monopoly pricing..."

Testing a single model of κ

Put the η_{jt} on the RHS and test whether $\lambda = 1$:

$$p_{jt} = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}, \theta_3) + \lambda \cdot \eta_{jt}\left(\mathbf{p}, \mathbf{s}, \theta_2, \kappa\right) + \omega_{jt} \text{ with } \mathbb{E}[\omega_{jt}|\mathsf{x}_t, \mathsf{w}_t, z_t] = 0$$

- We are basically running 2SLS with IV for the endogenous η_{jt}
- "Informal" test of Villas Boas (2007): $\mathbb{E}[\omega_{jt}|\mathsf{x}_{jt}, \mathsf{w}_{jt}, \eta_{jt}] = 0$.
  - Considers different forms of $f(\cdot)$: linear, exponential, logarithmic.
  - WP discusses this more than Restud version.
- Pakes (2017) uses Wollman (2018) data and BLP IV $\mathbb{E}[\omega_{jt} \mid \mathsf{x}_{jt}, \mathsf{w}_{jt}, f(\mathsf{x}_{_j})] = 0$.
- $\lambda \neq 0$ is hard to interpret.

Table 1: Wollman & Pricing Equilibrium.

Taken from Pakes, 2017, *Journal of Industrial Economics.*

	Price	(S.E.)	Price	(S.E.)
Gross Weight	.36	(0.01)	.36	(.003)
Cab-over	.13	(0.01)	.13	(0.01)
Compact front	-.19	(0.04)	0.21	(0.03)
long cab	-.01	(0.04)	0.03	(0.03)
Wage	.08	(.003)	0.08	(.003)
$\widehat{Markup}$	.92	(0.31)	1.12	(0.22)
Time dummies?	No	n.r.	Yes	n.r.
R^2	0.86	n.r.	0.94	n.r.

Note. There are 1,777 observations; 16 firms over the period 1992-2012. S.E.=Standard error.

Single Model Regressions

These are somewhat reassuring:

- $\lambda \approx 1$ for multiproduct-oligopoly
- Fit is pretty good $R^2 > 0.8$ and $R^2 > 0.5$ for within vehicle regressions (not shown).
- As a behavioral model, multiproduct demand estimation seems successful.
- But, do we know that an alternative $\mathcal{H}(\kappa)$ would have a $\lambda \neq 1$ or a lower R^2, and if so how low before we can "reject" the model?

Another idea (Bonnet and Dubois, Rand 2010) runs the following regression:

$$\log\left(p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \hat{\theta}_2, \kappa)\right) = h_s(\mathrm{x}_{jt}, \mathrm{w}_{jt}, \theta_3) + \omega_{jt}$$

- Run a regression for each κ and obtain $Q(\kappa) = \sum_{jt} \hat{\omega}_{jt}^2$
- Employ the non nested test of Rivers and Vuong (2002). Why?
- Working out the distribution of $Q(\kappa_1) - Q(\kappa_2) = T(\kappa_1, \kappa_2)$ is the hard part.
- Also this is OLS (or NLLS) and there are no instruments or exclusion restrictions for the supply side. Presumably we could add some and do GMM? (I think this is the "formal" test of Villas Boas (ReStud 2007)).

Villas Boas (2007)

TABLE 7

Percentage price-cost margins (PCM) by scenario

Wholesale, retail, and total PCM	Median	S.D.	Min	Max
Given random coefficient demand				
Model 1: Simple linear pricing: T-PCM	43·8	25·4	25·5	136·5
Model 2: Hybrid model: T-PCM	41·6	28·6	15·8	136·7
Model 3.1: Zero wholesale margin: R-PCM	21·1	9·0	12·5	50·0
Model 3.2: Zero retail margin: W-PCM	20·6	4·7	12·8	45·8
Model 4: Wholesale collusion: T-PCM	72·8	22·4	38·8	253·0
Model 5: Retail collusion: T-PCM	69·9	25·8	23·4	263·2
Model 6: Monopolist: T-PCM	40·6	11·5	21·8	103·7

PCM $= (p - c)/p$ where p is price and c is marginal cost. S.D., standard deviation; W, wholesale; R, retail; T, total.

Source: My calculations.

TABLE 8

Sample statistics of recovered costs and informal testing

	Recovered retail and wholesale costs				
	Mean	S.D.	Min	Max	Percentage <0
Model 1: Simple linear pricing	0·2730	0·1552	−0·1532	0·5364	1·35
Model 2: Hybrid model	0·2757	0·1681	−0·1541	0·5364	1·45
Model 3.1: Zero wholesale margin	0·3796	0·0907	0·1201	0·6298	0
Model 3.2: Zero retail margin	0·3815	0·0895	0·1302	0·6279	0
Model 4: Wholesale collusion	0·1352	0·1114	−0·4590	0·4409	13·12
Model 5: Retail collusion	0·1530	0·1329	−0·4895	0·5053	12·6
Model 6: Monopolist	0·2953	0·0951	−0·0089	0·5628	0·08

Notes: Recovered costs = p − epcm where p is retail price and epcm are the estimated margins. Last column has the percentage of cases with recovered estimated costs being negative.

Source: My calculations.

- Try out different models of price setting behavior, compute η markups.
- Unsurprisingly models with higher markups also have more costs $MC < 0$.
- Is this evidence of incorrect conduct assumption or inelastic demand?

Villas Boas (2007)

TABLE 10

p-Values for pairwise non-nested comparisons

	Alternative models						
H_0 model	1	2	3·1	3·2	4	5	6
1: Simple linear pricing	–	0·50	0·00	0·50	0·24	0·00	0·50
2: Hybrid	0·00	–	0·50	0·50	0·12	0·00	0·50
3.1: Zero wholesale margin	0·41	0·29	–	0·05	0·50	0·39	0·07
3.2: Zero retail margin	0·39	0·40	0·05	–	0·50	0·39	0·17
4: Wholesale collusion	0·49	0·48	0·50	0·50	–	0·48	0·50
5: Retail collusion	0·00	0·00	0·50	0·50	0·22	–	0·50
6: Monopolist	0·34	0·35	0·17	0·31	0·48	0·34	–
Chain size weighted							
1: Simple linear pricing	–	0·08	0·01	0·06	0·08	0·00	0·00
2: Hybrid	0·17	–	0·15	0·22	0·00	0·06	0·14
3.1: Zero wholesale margin	0·08	0·15	–	0·11	0·15	0·12	0·00
3.2: Zero retail margin	0·01	0·07	0·00	–	0·09	0·01	0·00
4: Wholesale collusion	0·00	0·05	0·04	0·09	–	0·00	0·02
5: Retail collusion	0·00	0·02	0·03	0·11	0·02	–	0·00
6: Monopolist	0·10	0·20	0·00	0·15	0·20	0·14	–

Notes: *p*-Values reported from non-nested, Cox-type (Smith, 1992) test statistics of the null model in a row being true against the specified alternative model in a column. Bottom part is a robustness check. It has the same format as above, but the non-nested comparisons are based on estimates for the case when the portion of the manufacturer's profit due to each retailer is weighted by the retailer's chain size.

Source: My calculations.

- Cox test has pairwise rejection problem (4) rejects (5) and (5) rejects (4).
- Likewise for 3.1 and 3.2 in top panel.

Recap

So far three approaches to exploit $\mathbb{E}[\omega_{jt}|\mathsf{x}_t, \mathsf{w}_t, z_t] = 0$

1. Put the markup on RHS and instrument for it to test $\lambda = 1$ (Wald)

$$p_{jt} = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}, \theta_3) + \lambda \cdot \eta_{jt}\left(\mathbf{p}, \mathbf{s}, \widehat{\theta_2}, \kappa\right) + \omega_{jt}$$

2. Put the markup on LHS assuming $\lambda = 1$ and test goodness of fit of supply equation (Anderson Rubin)

$$p_{jt} - \eta_{jt}\left(\mathbf{p}, \mathbf{s}, \widehat{\theta_2}, \kappa\right) = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}, \theta_3) + \omega_{jt}$$

3. Estimate supply and demand simultaneously $[\theta_1, \theta_2, \theta_3]$ and compare goodness of fit for different κ. (Likelihood Ratio)

Which Test? Duarte, Magnolfi, Sølvsten, Sullivan (QE 2024)

Take the supply equation seriously as a moment condition and ask which classical test to run on a menu of markups $\{\eta^m\}$:

- **Specification tests** of one model at a time (Wald on $\lambda = 1$, Anderson–Rubin / GMM-J on $\mathbb{E}[\omega^m z] = 0$, Cox): with enough data every model is rejected, because no conduct model is exactly right. Villas-Boas's Cox test rejecting in both directions is the symptom.
- **Model selection** (Rivers–Vuong 2002): compare two models' GMM criteria, $Q_W(\eta^{m_1}) - Q_W(\eta^{m_2})$, under the null that they are equally misspecified. Only RV behaves sensibly when both models are wrong, which is always.
- The catch: RV is degenerate when neither model is falsified through the instruments (degeneracy only at the origin, their Proposition 3). DMSS define weak instruments for testing, extend Stock–Yogo to model selection, and give an effective-F diagnostic with critical values for worst-case size and best-case power (`pyRVtest`).
- Their criterion is a finite-dimensional 2SLS quadratic form, $\bar{\mathbf{m}}_m'(z'z)^{-1}\bar{\mathbf{m}}_m$ with $\bar{\mathbf{m}}_m = \frac{1}{n}\sum_i z_i\omega_i^m$ (a K-vector of unconditional moments). Keep that in mind: the two BCS criteria below are a single optimal direction and the full conditional-moment norm, and the DMSS object is the finite-K thing in between.

Model Selection under Misspecification

What Is the Null? Ranking Misspecified Models

Each conduct m implies markups η_{jt}^m, hence an implied cost and a cost residual:

$$c_{jt}^m = p_{jt} - \eta_{jt}^m = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \omega_{jt}^m, \qquad \mu_m(z) \equiv \mathbb{E}[\omega_{jt}^m \mid z_{jt}], \qquad Q(\eta^m) \equiv \|\mu_m\|^2 = \mathbb{E}[\mu_m(z)^2] \geq 0.$$

- Correct conduct: $\mu_m \equiv 0$ and $Q(\eta^m) = 0$. On real data every $Q(\eta^m) > 0$: no model on the menu satisfies the restriction exactly.
- So the honest hypothesis is not "Bertrand is true" but Vuong (1989) / Rivers–Vuong (2002) / Chen–Hong–Shum (2007):

$$H_0 : Q(\eta^{m_1}) = Q(\eta^{m_2}) \quad \text{(equally wrong)} \qquad \text{vs.} \qquad H_A : \text{one is closer to the moment condition.}$$

- A "test of conduct" is therefore a ranking: which markups are closest to $\mathbb{E}[\omega \mid z] = 0$. "Model m is in the model confidence set" means it is the hardest to beat, not that it is correct.
- The cost function h_s is a nuisance; it is the pseudo-true cost function under each (possibly wrong) conduct, well defined because the criterion is shared across models.

Degeneracy: When Neither Model Is Falsified

The RV statistic is $\sqrt{n}\,(\widehat{Q}_1 - \widehat{Q}_2)/\widehat{\sigma}$. Write $\omega^m = \mu_m(z) + \varepsilon^m$. The influence of the contrast is $(\mu_{m_1}^2 - \mu_{m_2}^2) + 2(\mu_{m_1}\varepsilon^{m_1} - \mu_{m_2}\varepsilon^{m_2})$, so

$$\sigma_{\text{lin}}^2 = \text{Var}(\mu_{m_1}^2 - \mu_{m_2}^2) + 4\,\mathbb{E}\big[\text{Var}(\mu_{m_1}\varepsilon^{m_1} - \mu_{m_2}\varepsilon^{m_2} \mid z)\big].$$

- $\sigma_{\text{lin}}^2 = 0$ only when $\mu_{m_1} \equiv \mu_{m_2} \equiv 0$ on the span of the instruments: neither model is falsified. Both correct, or z irrelevant for both: DMSS's weak instruments for testing ("degeneracy only at the origin"). There $\widehat{\sigma} \to 0$ and the normal approximation fails.
- Equal conditional moments with a common misfit, $\mu_{m_1} \equiv \mu_{m_2} \not\equiv 0$, are not degenerate: $\sigma_{\text{lin}}^2 = 4\mathbb{E}[\mu^2\,\text{Var}(\Delta\eta \mid z)] > 0$. The null is simply true, there is nothing to rank, and the statistic stays normal. (Unlike the likelihood case of Vuong / Chen–Hong–Shum, where equal densities zero out every observation's contribution.)
- Nearly identical markups are not degeneracy either: scale $\Delta\eta$ by $\delta \to 0$ and the signal and σ_{lin} both scale with δ, so the regime of the studentized statistic does not move. What falls is power.
- What governs the regime is instrument strength for testing (DMSS Proposition 4): the concentration parameter of $\Delta\eta$ on the excluded z net of controls, plus that of the level $\mu_{m_1} + \mu_{m_2}$. Weak-instrument comparisons sit near the boundary; the rest of the lecture is about inference that survives there.

A Different Degeneracy: Constant-Markup Conducts Are a Straw Man

Suppose conduct m implies a constant proportional markup, $p = \phi \cdot mc$ (perfect competition is $\phi = 1$; symmetric CES-type pricing is $\phi > 1$). Then

$$\omega_\phi = p/\phi - h_s(\mathsf{x}, \mathsf{w}) = \tfrac{1}{\phi}\,\varrho, \qquad \varrho \equiv \text{the price residual on } (\mathsf{x}, \mathsf{w}), \qquad Q(\phi) = \tfrac{1}{\phi^2}\,\|\mathbb{E}[\varrho \mid z]\|^2.$$

- The moments are proportional, $\mu_\phi = \mathbb{E}[\varrho \mid z]/\phi$, so $\Delta\widetilde{Q} = \|\mathbb{E}[\varrho \mid z]\|^2(\phi_1^{-2} - \phi_2^{-2})$: T_{RV} is normal with power and consistently selects the larger markup. Scale, not fit; no instrument fixes it (lossless).
- So "we reject perfect competition" means only that the price residual projects onto z. It is a statement about demand-side variation in prices, not about conduct.
- Rules: never compare two constant-markup conducts; treat perfect competition as a sanity check, not an informative comparison. Repairs: outside information on the markup *level* (Nevo 2001), or studentizing each moment, which leaves nothing to test.
- The informative comparisons are between conducts that imply different patterns of markups across products and markets: single-product vs. own-firm vs. common ownership vs. monopoly. Those differ in how much of the cross-price structure the firm internalizes.

Backus, Conlon, Sinkinson: the scalar test (BCS 1.0)

Basic Setup

We start with marginal revenue and marginal cost (unobserved ω, observed $h_s(\cdot)$)

$$mr_{jt}^m = mc_{jt}$$
$$p_{jt} - \eta_{jt}^m = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \omega_{jt}^m$$

- Let's be vague/flexible with $h_s(\cdot)$ for now, but I don't know the production function.
- Assume: Demand and hence η_{jt}^m are known (given conduct).
- Idea (η^{m_1}, η^{m_2}) are own-profit vs. common ownership, or single-product vs. own-profit, or Cournot vs. Bertrand.

The Question

Two competing markups $(\eta_{jt}^{m_1}, \eta_{jt}^{m_2})$: which fits the data better?
(both may be misspecified)

$$p_{jt} = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \tau\, \eta_{jt}^{m_1} + (1-\tau)\, \eta_{jt}^{m_2} + \omega_{jt}$$

Model is defined by a conditional moment restriction $\mathbb{E}[\omega_{jt}|z_{jt}] = 0$

- $\tau = 1$ is m_1, $\tau = 0$ is m_2. The null is equal fit, $\widetilde{Q}(\eta^{m_1}) = \widetilde{Q}(\eta^{m_2})$, not "$\tau = 1$": both may be wrong.
- This is a model selection problem or a non nested testing problem.
  - We might want to compare more than two alternatives (too bad).
- Obvious endogeneity problem with η_{jt}!

Our Idea: Motivation #2 (Misspecification)

Index the true model by m_0 and a wrong one by m_1. Then,

$$p_{jt} - \eta_{jt}^{m_0} = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \omega_{jt}^{m_0}.$$

To motivate a useful test, we ask what happens when we estimate supply with the wrong conduct model m_1:

$$p_{jt} - \eta_{jt}^{m_1} = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \underbrace{\underbrace{\eta_{jt}^{m_0} - \eta_{jt}^{m_1}}_{\equiv \Delta\eta_{jt}} + \omega_{jt}^{m_0}}_{\omega_{jt}^{m_1}}.$$

- Misspecifying conduct introduces an omitted variable: the difference in markups.
- Our test is premised on detection of this omitted variable.

The Instrument: Optimal Among Single Directions (BCS 1.0)

Collapse the moment against one direction $b(z)$: $\bar{m}_m(b) = \mathbb{E}[\omega^m b(z)]$ and $\widetilde{Q}(\eta^m) = \bar{m}_m^2$. Which b discriminates m_1 from m_2 best?

$$\omega^{m_1} - \omega^{m_2} = -(\Delta\eta - \mathbb{E}[\Delta\eta \mid \mathsf{x}, \mathsf{w}]) \quad \Rightarrow \quad g(z) \equiv \mathbb{E}[\Delta\eta \mid \mathsf{x}, \mathsf{w}, z] - \mathbb{E}[\Delta\eta \mid \mathsf{x}, \mathsf{w}]$$

- g is the projection of the markup difference onto the excluded instruments after the cost controls are partialled out (Frisch–Waugh–Lovell / Robinson). It is the Chamberlain optimal instrument for the omitted variable $\Delta\eta$ of the previous slide.
- **Optimality:** among all single directions, g uniquely maximizes the signal-to-noise ratio for telling the two conducts apart (Cauchy–Schwarz; heteroskedasticity adds a $1/\sigma^2(z)$ weight), attaining the full conditional-moment distance $\|\mu_{m_1} - \mu_{m_2}\|$. RV on it is lossless in population: with $\|g\| = 1$, $\bar{m}_1^2 - \bar{m}_2^2 = \|\mu_{m_1}\|^2 - \|\mu_{m_2}\|^2$, the full-CMR ranking and null.
- **Failure set shrinks:** a generic b also fails when $b \perp \mu_{m_1} - \mu_{m_2}$ (spurious equal fit) or $b \perp \operatorname{span}\{\mu_{m_1}, \mu_{m_2}\}$ (Vuong degeneracy); along g both collapse to equal conditional moments, $d = 0$, where g itself vanishes. With $\mu_{m_1} \equiv \mu_{m_2} \equiv 0$ that is the DMSS degeneracy; with a common misfit $\hat{g}$ is first-stage noise, the limit is non-degenerate, but the fixed-$\hat{g}$ SE is too small (9–15% rejection at nominal 5%).
- **The subtraction is not redundant** even though $(\mathsf{x}, \mathsf{w}) \subset z$: it is what makes $\mathbb{E}[g \mid \mathsf{x}, \mathsf{w}] = 0$, and that is what makes the moment insensitive to errors in h_s (next slide).
- Practical: $\hat{g}$ is two cross-fit regressions of $\Delta\eta$ (random forest, boosting, or a sieve), one on everything and one on the controls, differenced.

Orthogonality and Cross-Fitting in One Slide

Two nuisance functions, h_s and g, are estimated and plugged in. Why does their estimation error not contaminate the test? (Chernozhukov et al. 2018 double machine learning, applied to a model-selection statistic.)

- **Cost direction.** Perturb $h_s \to h_s + \delta h$: the moment moves by $-\mathbb{E}[\delta h(\mathsf{x}, \mathsf{w}) \cdot \mathbb{E}[g \mid \mathsf{x}, \mathsf{w}]] = 0$ by construction of g. First-order error in the cost function cannot enter. (This is Neyman orthogonality, delivered by the FWL subtraction.)
- **Instrument direction.** Perturb g: the moment moves by $\mathbb{E}[\mu_m \delta g]$, zero for every δg iff $\mu_m \equiv 0$. No null is needed: a model that satisfies the CMR has its moment pinned at zero however badly g is estimated, so with a true model on the menu $\hat{g}$-error moves power, not size. In the both-wrong null ($\bar{m}_1 = -\bar{m}_2 \neq 0$) a first-order term survives along the level direction $\mu_{m_1} + \mu_{m_2}$, and any first stage leaves it out of the fixed-$\hat{g}$ SE: $\mathrm{Var}(\mathsf{T}_{RV}) \to 1 + R_r^2\,(1 - R_{fs}^2)/R_{fs}^2$, with R_{fs}^2 the first-stage R^2 of $\Delta\eta$ on the excluded z net of controls and R_r^2 the share of the level they explain. In cereal under the optimal IV, $R_{fs}^2 \approx 0.05$–0.10, so the fixed-$\hat{g}$ T is overstated by a factor of about 2. Report R_{fs}^2; a refit bootstrap or generated-instrument correction restores the term; the full-CMR criterion has no estimated direction and is immune.
- **Cross-fitting.** Split the clusters (whole markets, never products within a market) into K folds; evaluate each observation at nuisance functions fit on the other folds. This removes the own-observation bias that in-sample fitting puts into a squared moment ($\mathbb{E}[\widetilde{Q}] = Q + \mathrm{Var}(\bar{m}) +$ overfit), without complexity restrictions on the learner.
- **Rate.** Remaining bias is a product of two nuisance errors, negligible once each converges faster than $n^{-1/4}$ (Bickel–Ritov). Random forests and boosting on a few hundred thousand rows clear this comfortably.

Inference for the Scalar Test

Aggregate the integrand to cluster totals $S_c^m = \sum_{i \in c} w_i \, \hat{\omega}_i^m \hat{g}_i$, the cluster being a union of whole markets (cereal scalar tables: chain×year, $C = 242$), so $\bar{m}_m = \frac{1}{n} \sum_c S_c^m$ and $\Delta\widetilde{Q} = \bar{m}_1^2 - \bar{m}_2^2$.

- **Analytic (delta method).** $\mathsf{T}_{RV} = \sqrt{n}\,\Delta\widetilde{Q}/\hat{\sigma}$ with $\hat{\sigma}^2$ from the cluster totals and $\nabla f = (2\bar{m}_1, -2\bar{m}_2)$. Valid away from the boundary; as $\bar{m}_m \to 0$, $\hat{\sigma} \to 0$ and it fails.
- **Bootstrap 1, multiplier on the influence function.** Rademacher weights e_c on the linearized contribution: reproduces the analytic SE, same boundary problem (Kline–Santos 2012 score bootstrap).
- **Bootstrap 2, score-level wild.** Perturb each model's moment before squaring, $\bar{m}_m^* = \bar{m}_m + \frac{1}{n} \sum_c e_c \tilde{S}_c^m$, then square. Keeps the cross term *and* the quadratic term, so it self-adapts: normal far from the boundary, fat-tailed near it, with no pretest. The variance it adds relative to the linear term scales as $1/\mathsf{T}_m^2$, the per-model moment t-ratio.
- **Bootstrap 3, wild cluster bootstrap-t.** Studentize every draw; the few-cluster refinement (Cameron–Gelbach–Miller 2008).
- Because a squared mean is biased upward, Bootstraps 2–3 recenter: report $\Delta\widetilde{Q} - (\widehat{V}_1 - \widehat{V}_2)$ against the centered bootstrap law.
- No refits: every bootstrap only reweights cached cluster totals. Markups and first stages are computed once.

Algorithm (BCS 1.0)

For a pair (m_1, m_2); the menu test repeats over all pairs.

1. **Split.** Partition the C clusters into K folds (3 in cereal); every observation inherits its cluster's fold. The same split is reused for every first stage.
2. **Cost residuals, per model.** For each m and fold k, fit $\hat{h}_s^{(-k)}(\mathsf{x}, \mathsf{w}) = \hat{\mathbb{E}}[MC^m \mid \mathsf{x}, \mathsf{w}]$ on the other folds and residualize on fold k; fixed effects (product, chain, year) absorbed by a within-transform first. Computed once per model and reused in every pair.
3. **Instrument, per pair.** Cross-fit the two projections of $\Delta\eta$ and difference them: $\hat{g} = \hat{\mathbb{E}}^{(-k)}[\Delta\eta \mid \mathsf{x}, \mathsf{w}, z] - \hat{\mathbb{E}}^{(-k)}[\Delta\eta \mid \mathsf{x}, \mathsf{w}]$.
4. **Moments.** Cluster totals S_c^m, sample moments $\bar{m}_m$, criteria $\widetilde{Q}_m = \bar{m}_m^2$, contrast $\Delta\widetilde{Q}$.
5. **Inference.** T_{RV}: analytic cluster SE by default, Bootstrap 2 or 3 near the boundary, cached totals only.
6. **Menu.** Pairwise matrix of contrasts $\rightarrow$ model confidence set: the models never significantly beaten.

BCS in one slide

Two competing markups (markdowns) $(\eta_{jt}^{m_1}, \eta_{jt}^{m_2})$. Which fit data better? (both may be misspecified)

$$p_{jt} = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \tau\, \eta_{jt}^{m_1} + (1-\tau)\, \eta_{jt}^{m_2} + \omega_{jt} \text{ where } \mathbb{E}[\omega_{jt} | \mathsf{x}_{jt}, \mathsf{w}_{jt}, z_{jt}] = 0$$

The only IV that matters is the predicted markup difference, net of the cost controls, which we get from two first stages

$$g(z_{jt}) = \mathbb{E}[\eta_{jt}^{m_1} - \eta_{jt}^{m_2} \mid \mathsf{x}_{jt}, \mathsf{w}_{jt}, z_{jt}] - \mathbb{E}[\eta_{jt}^{m_1} - \eta_{jt}^{m_2} \mid \mathsf{x}_{jt}, \mathsf{w}_{jt}]$$

As long as we have some exclusion restrictions z_{jt} that:

- **Relevance**: Predict markups beyond the cost shifters $(\mathsf{x}_{jt}, \mathsf{w}_{jt})$.
- **Exclusion**: Do not affect marginal cost.

Rest of paper: make $h(\cdot)$ and $g(\cdot)$ as flexible as possible (cross-fit), partial the controls out of g, and get a Vuong test whose inference survives the degeneracy boundary.

BCS in one slide

Two competing markups (markdowns) $(\eta_{jt}^{m_1}, \eta_{jt}^{m_2})$. Which fit data better? (both may be misspecified)

$$p_{jt} - \eta_{jt}^{m_1} = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \omega_{jt} \text{ if } \tau = 1 \ (m_1)$$
$$p_{jt} - \eta_{jt}^{m_2} = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \omega_{jt} \text{ if } \tau = 0 \ (m_2)$$

The only IV that matters is the predicted markup difference, net of the cost controls, which we get from two first stages

$$g(z_{jt}) = \mathbb{E}[\eta_{jt}^{m_1} - \eta_{jt}^{m_2} \mid \mathsf{x}_{jt}, \mathsf{w}_{jt}, z_{jt}] - \mathbb{E}[\eta_{jt}^{m_1} - \eta_{jt}^{m_2} \mid \mathsf{x}_{jt}, \mathsf{w}_{jt}]$$

As long as we have some exclusion restrictions z_{jt} that:

- **Relevance**: Predict markups beyond the cost shifters $(\mathsf{x}_{jt}, \mathsf{w}_{jt})$.
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Rest of paper: make $h(\cdot)$ and $g(\cdot)$ as flexible as possible (cross-fit), partial the controls out of g, and get a Vuong test whose inference survives the degeneracy boundary.

The full conditional-moment test (BCS 2.0)

The Full Conditional-Moment Test in One Slide (BCS 2.0)

Use all of $\mu_m(z) = \mathbb{E}[\omega^m \mid z]$, not one direction: $Q(\eta^m) = \|\mu_m\|^2$, $H_0 : Q(\eta^{m_1}) = Q(\eta^{m_2})$. The scalar BCS moment, the DMSS 2SLS quadratic form, and this norm are one direction, K directions, and all directions of the same object.

$$\widehat{Q}(\eta^m) = \frac{1}{C(C-1)} \sum_{c \neq c'} (u_c^m)' u_{c'}^m, \qquad u_c^m = \widehat{\Sigma}^{-1/2} \sum_{j \in c} \widehat{\omega}_j^m \, b(z_j), \qquad b \text{ mean-centered.}$$

- **Cross-cluster U-statistic**: products of residuals from different markets are unbiased for $\|\mu_m\|^2$; the own-cluster terms carry the estimation noise that inflates any plug-in (the "positive shift"), so they are deleted. Centering b removes an $O(1)$ bias from the intercept of h_s (FWL).
- **Why bother**: the degenerate set is the same as the scalar test's (neither model falsified), so it buys no power there. What it buys is an instrument side with no estimated direction: where the scalar $\hat{g}$ is first-stage noise (equal conditional moments, the both-wrong null) the norm keeps nominal size. One cross-fit regression per model, nothing refit per pair.
- **The price**: under a correct model the criterion sits at its minimum and is not normal, so inference has two regimes (two slides on).

The Full-CMR Procedure: Which Regressions, and How the Criterion Is Formed

For a menu of models m; only step 2 is per model, nothing is refit per pair.

1. **Split.** Partition the C clusters (whole markets) into K folds; the same split serves every regression below.
2. **Cost residuals, per model.** Regress $MC^m = p - \eta^m$ on the cost controls (x, w) with the FE absorbed by a within-transform, out of fold: $\widehat{\omega}_i^m = MC_i^m - \widehat{h}_s^{(-k)}(\mathrm{x}_i, \mathrm{w}_i)$. The same residuals as the scalar test; the only nonparametric nuisance.
3. **Instrument basis.** Take a series basis $b(z) \in \mathbb{R}^{K_z}$ on the excluded instruments (a low-order polynomial). Regress every column on $(\mathrm{x}, \mathrm{w}, \mathrm{FE})$, cross-fit, and keep the residuals $\tilde{b}(z)$: the controls are partialled out of the basis exactly as they are out of $\hat{g}$ (mean-centering is the constant-only case). Whiten with $\widehat{\Sigma} = \frac{1}{n}\sum_i \tilde{b}_i \tilde{b}_i'$.
4. **The CMR estimate is a regression.** Regress $\widehat{\omega}^m$ on $\tilde{b}(z)$: $\widehat{\mu}_m(z) = \tilde{b}(z)' \, \widehat{\Sigma}^{-1} \, \frac{1}{n}\sum_i \tilde{b}_i \widehat{\omega}_i^m$, the fitted value. Its mean square, $\hat{\nu}_m' \hat{\nu}_m$ with $\hat{\nu}_m = \widehat{\Sigma}^{-1/2} \frac{1}{n}\sum_i \tilde{b}_i \widehat{\omega}_i^m$, is the 2SLS quadratic form of DMSS. It is biased upward by the estimation variance of $\widehat{\mu}_m$.
5. **Debias with cluster scores.** $G_c^m = \sum_{i \in c} w_i \, \widehat{\omega}_i^m \tilde{b}_i$, $u_c^m = \widehat{\Sigma}^{-1/2} G_c^m$, and $\widehat{Q}(\eta^m) = \frac{1}{C(C-1)}\left(\| \sum_c u_c^m \|^2 - \sum_c \| u_c^m \|^2\right)$: the cross-cluster products only. Dropping $c = c'$ drops $\mathbb{E}\|\widehat{\mu}_m - \mu_m\|^2$.
6. **Contrast and inference.** $\Delta\widehat{Q} = \widehat{Q}(\eta^{m_1}) - \widehat{Q}(\eta^{m_2})$; per-cluster influence $\widehat{\psi}_c$ (leave-one-out $\hat{\nu}$, weight-matrix term); unified multiplier bootstrap on the cached u_c^m; p-value from the bootstrap law (next slide).
7. **Menu.** Pairwise matrix of contrasts $\rightarrow$ model confidence set.

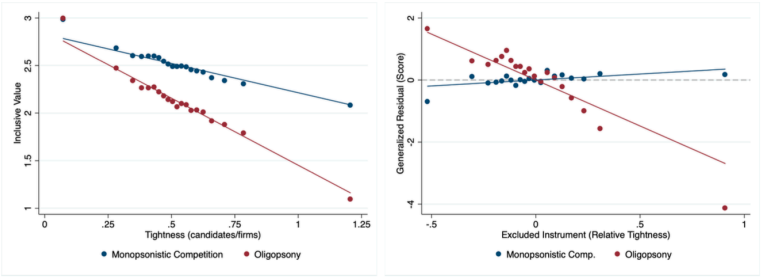
Inference in One Slide: Two Regimes, One Bootstrap

$$\Delta\widehat{Q} - (Q_{m_1} - Q_{m_2}) = \underbrace{\frac{1}{C}\sum_c \psi_c}_{\sqrt{C}\text{-normal if } \sigma^2_{\text{lin}}>0} + \underbrace{\frac{1}{C(C-1)}\sum_{c\neq c'} \Psi(c,c')}_{\text{rate-}C\text{ weighted }\chi^2 \text{ if } \mu_{m_1}\equiv\mu_{m_2}\equiv 0}$$

- **Non-degenerate**: at least one model violates the CMR on the span of z; cluster-robust RV is fine. **Degenerate**: neither does (both correct, or weak instruments for testing); the linear term vanishes, $C\widehat{Q} \to \sum_r v_r(\chi^2_{1,r} - 1)$, mean zero and right-skewed, so a normal cutoff under-covers the upper tail. The standard error differs by $\sqrt{C}$ between regimes and weak-instrument comparisons sit near the boundary. Because $\widehat{\Sigma}$ is estimated, ψ_c carries a weight-matrix term: per observation the influence of Q_m is $2v\omega^m - v^2 - Q_m$ with v the fitted moment (DMSS eq. 8); dropping v^2 overstates the variance.
- **One multiplier bootstrap** on both terms, $\Delta\widehat{Q}^* - \Delta\widehat{Q} = \frac{1}{C}\sum_c e_c\widehat{\psi}_c + \frac{1}{C(C-1)}\sum_{c\neq c'} e_c e_{c'}\widehat{\Psi}(c,c')$, one Rademacher weight per cluster shared across terms and models. It nests the linear RV and the score-wild bootstrap and interpolates between them with no pretest and no switch; the p-value comes from the bootstrap law, not a normal table; nothing is refit.
- **Diagnostics**: $\widehat{F} = \max_m C\widehat{Q}(\eta^m)$ over the pair (near degeneracy only when both criteria are near zero) and the first-stage fit of $\Delta\eta$ on z (is there anything to rank?), both descriptive. **Open**: uniform validity across the boundary (the Shi 2015 analogue), demand-estimation uncertainty, a Hansen model confidence set.

Examples

Figure 5: Vuong Test



(a) First Stage **(b)** Visualizing the Vuong Test

Note: Panel (a) plots the "first stage" relationship between the model-implied inclusive values Λ_i and Λ_i^{-j} and the instrumental variable t_{ij}, conditional on firm covariates z_j and candidate covariates x_i and two-week period dummies. Panel (b) plots the relationship between generalized residuals and the excluded instrument for the non-predictive monopsonistic competition and oligopsony models. Under proper specification, the correlation of the generalized residuals and the excluded instrument should be zero (the dashed line). The larger the deviation from zero, the greater the degree of mis-specification of the model.

- Offered wages for online job platform
- Compares Monopsony vs. oligopolistic competition vs. perfect comp.
- Compares tailored offers vs. not (price discrimination).

Scuderi JMP 2022: Models of Labor Supply

- Firms ignore competitors (Monopsony)
- Firms offer wages independent of candidate characteristics (experience, demographics).
- Firms are definitely NOT paying MPL.

Table 4: Non-Nested Model Comparison Tests (Rivers and Vuong, 2002)

Model	(1) Monopsonistic Comp.	(2) Monopsonistic Comp.	(3) Oligopsony	(4) Oligopsony
	Not Predictive	Type Predictive	Not Predictive	Type Predictive
Perfect Competition	-54.84	-54.40	-39.92	-39.91
Monopsonistic, Not Predictive	–	7.83	3.98	2.69
Monopsonistic, Type Predictive		–	2.77	1.54
Oligopsony, Not Predictive			–	-3.67
Oligopsony, Type Predictive				–

Note: This table reports test statistics from the Rivers and Vuong (2002) non-nested model comparison procedure. Positive values imply the row model is preferred to the column model. Under the null of model equivalence, the test statistics are asymptotically normal with mean zero and unit variance.

specification. In the figure, the generalized residuals for the monopsonsitic competition alternative are closely aligned with the x-axis, while the generalized residuals for the oligopsony alternative are strongly negatively related to tightness.

Our tests therefore suggest that models of firm behavior in which firms both ignore strategic interactions in wage setting and do not tailor wage offers to candidates on the basis of predictable preference variation are closer approximations to firms' true bidding behavior on the platform than are models in which firms act strategically and tailor offers. In Appendix F, we report the testing results using the original Vuong (1989) likelihood comparison test, which yield qualitatively identical model comparisons. In the following analysis, we adopt the not-predictive monopsonostic competition model as our preferred model of conduct.

Starc Wollmann: Generic Pharma Cartel + Entry

Do cartels encourage entry with high prices? Starc and Wollmann (2022) apply the BCS test to the generic-drug cartel; the menu is competitive vs. joint-profit-maximizing pricing before and after cartel formation, with non-members best responding.

> *in 2013, Teva Pharmaceuticals, the largest generic firm, hired NP, a marketing executive with especially strong industry relationships, and tasked her with "price increase implementation."1 Over an 18-month period, industry participants exchanged thousands of calls and texts—alongside countless LinkedIn, Facebook, and WhatsApp messages and face-to-face conversations—with contacts at rival firms to coordinate the increases (Complaint, page 322).2 Following this period, prescription drug expenditures by governments, private insurers, and individuals rose sharply by billions of dollars.*

Starc Wollmann: Testing Results (prices rise with the cartel, less in markets that invite entry)

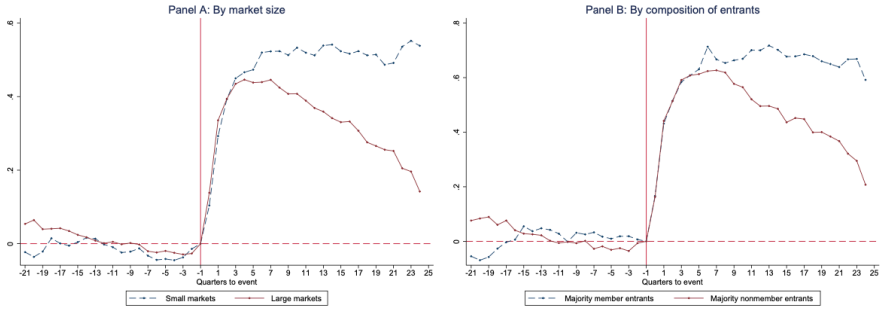
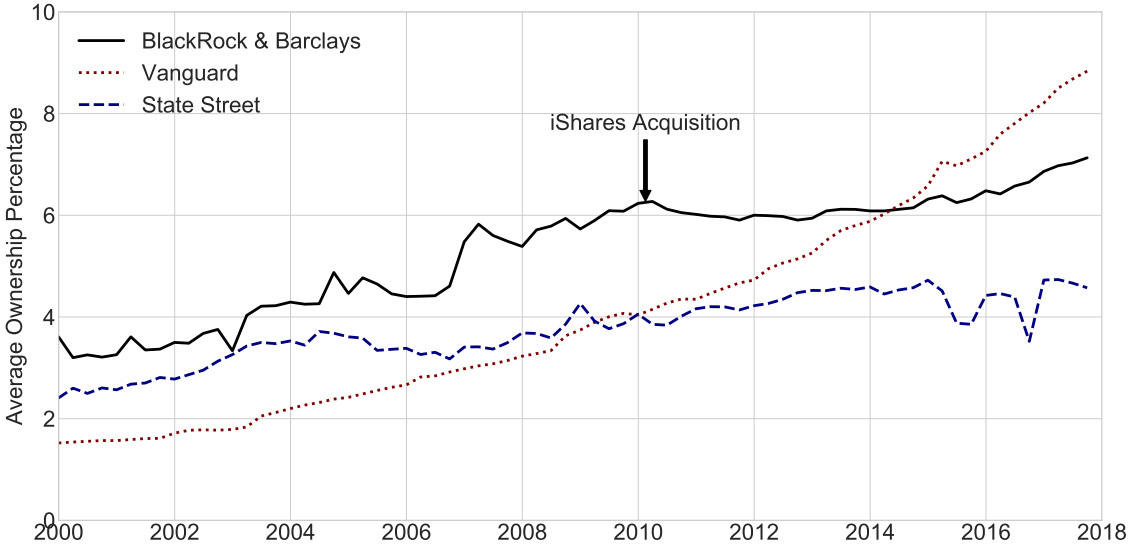


Figure III: *Prices after cartel formation*

In this figure, we plot estimates of β^{τ}, which are derived from equation 1, against x^{τ}, which represents event time in quarters. For Panel A, we proxy for market size by computing the number of prescriptions filled in the period just prior to NP joining Teva, and we distinguish between large and small cartelized markets based on how this figure relates to the median value. For Panel B, we calculate the fraction of entrants that arrive as a result of ANDAs filed after NP joins Teva and identify markets in which a majority of those firms are members and nonmembers, respectively.

Application: Common Ownership (Backus Conlon Sinkinson)

Rise of the Big Three (S&P 500 components, BCS AEJ:Micro 2021)



The Reduced-Form Literature (and its Critics)

Symmetric homogeneous-good Cournot with profit weights gives a share-weighted average markup (with market demand elasticity ϵ):

$$\sum_f s_f \cdot \frac{p - mc_f}{p} = \frac{1}{\epsilon} \underbrace{\mathbf{s}' \mathcal{H}(\kappa) \mathbf{s}}_{\equiv MHHI}, \qquad MHHI\Delta \equiv MHHI - \underbrace{\mathbf{s} \cdot \mathbf{s}}_{HHI}$$

Some regress

$$\log\left(p_{rjt}\right) = \beta_1 \cdot MHHI\Delta_{rt} + \beta_2 \cdot HHI_{rt} + \theta \cdot X_{rjt} + \alpha_t + \nu_{rj} + u_{rjt}$$

- Azar, Schmalz, Tecu (JF 2018): airline routes with more common ownership have 3–7% higher fares.
- But $MHHI\Delta$ is a function of equilibrium market shares – endogenous! (O'Brien Waehrer 2017). Price-concentration regressions all over again.
- Dennis, Gerardi, Schnitzler (JF 2022): not robust (driven by the share component); Kennedy, O'Brien, Song, Waehrer (2017): structural version finds nothing.
- Better idea: treat CO vs. own-profit maximization as a conduct testing problem.

The Menu Is an Internalization Ladder

In cereal the candidate conducts differ in how much of the cross-price structure the firm prices on:

$$\underbrace{\text{single-product}}_{\text{own elasticities only}} \prec \underbrace{\text{own-profit}}_{\text{Nash–Bertrand, own firm}} \prec \underbrace{\text{common ownership } \kappa}_{\text{own firm + profit weights on rivals}} \prec \underbrace{\text{monopoly}}_{\text{everything}}$$

- All of them, single-product included, price above marginal cost. The test ranks models by the recapture they internalize, not by whether market power exists.
- So "single-product fits the moment best" is a statement about cross-price internalization, not a rejection of market power, and not a statement that firms ignore their sister brands in any literal sense.
- Perfect competition is on the menu as a straw man only (the constant-markup degeneracy). Cournot vs. Bertrand would be a different ladder.
- This is the substantive content of a criterion $\|\mathbb{E}[\omega^m \mid z]\|^2$ here: it orders the ladder; it does not estimate a markup level. The markup level comes from demand.

Near-Collinear Markups Lower Power; Instrument Strength Sets the Regime

- Across the menu the implied markups are ≥ 0.92 correlated, so $\Delta\eta = \eta^{m_1} - \eta^{m_2}$ is small relative to η^m. That does not put the test near the degeneracy boundary: the scale of $\Delta\eta$ cancels from the studentized statistic. It lowers power, because the misfit a wrong model adds is small relative to the noise in the cost residual.
- What the test needs is that the excluded z predict $\Delta\eta$ net of the controls: instrument strength for conduct, a different object from strength for demand. An instrument can move prices and shares beautifully and project onto $\Delta\eta$ not at all.
- Monte Carlo with one model correct: rejection $1.00 \to 0.05$ as corr(η^{m_1}, η^{m_2}) rises from 0.97 to 0.999. That is power falling as the wrong model's violation shrinks, with both moments going to zero, not a degeneracy caused by collinear markups.
- Practical implication: before testing, regress $\Delta\eta$ on the instruments (net of controls) and report the fit, the effective first-stage F. If nothing explains the markup difference beyond the cost shifters, no test, scalar or norm, will rank the pair. This is the DMSS weak-instruments-for-testing diagnostic in plain clothes, and for the scalar test the same R^2_{fs} sets how far its fixed-$\hat{g}$ SE is off.

Validity Is Orthogonality to the Cost Shock, Not to Price

A conduct instrument must be excluded from ω; it must correlate with price, since a useful instrument moves markups. Choosing price-orthogonal instruments manufactures weak ones. The cereal instrument groups, after partialling $(\mathsf{x}, \mathsf{w}, \mathsf{FE})$ out:

Group	What it is	Strength for conduct
`opt_iv`	demand-side Chamberlain optimal IV, $\mathbb{E}[\partial\xi/\partial\theta \mid z]$ at a flexible $\widehat{\mathbb{E}}[p \mid z]$	strong (effective $F \approx 50$–60)
`demog`	chain-level income and demographic moments	moderate ($F \approx 10$–17); mostly absorbed by the FE
`opc`	price of a grain the product does *not* use (a rival's cost shifter)	valid; $\approx 10\%$ of its variance survives the FE ($F \approx 5$–12)
`diff`	BLP / Gandhi–Houde differentiation instruments	relevant ($F \approx 30$–60)

- The optimal IV is not circular: demand was estimated without a supply side, so $\widehat{\mathbb{E}}[p \mid z]$ is a reduced-form fit, not an equilibrium price under an assumed conduct. Its near-perfect span of expected prices is the source of its power.
- The BLP differentiation instruments are built for demand. Their strength for conduct is a separate empirical question, answered by the first-stage F for $\Delta\eta$ and not by their strength for demand.
- Partial the controls out of the instrument basis (FWL), as the scalar test partials $\hat{g}$; otherwise control overlap inflates the Gram matrix and groups are not comparable. Partialling sharpens valid strong instruments and exposes FE-collinear ones as weak.

Fixed Effects, Weighting, and the Specification of h_s

- Fixed effects in h_s absorb cost heterogeneity that would otherwise read as moment violation, and they absorb each instrument's FE-collinear variation: the chain-level demographics and the product-level grain prices lose most of theirs. Product fixed effects matter most.
- Example: across a 20-cell sweep (5 FE sets $\times$ 2 weightings $\times$ 2 instrument groups) the model confidence set is $\{\text{single}\}$ in every cell with product FE; without product FE and without weights the collusive models survive under the optimal IV.
- Servings weighting roughly halves the criterion magnitudes. Additive versus interacted FE can flip a pairwise decision; the full product$+$year$+$chain set is invariant.
- Lesson: the specification of h_s (what is a cost control, which FE, how flexible, weighted or not) is as much a part of the identification argument as the instruments. Report it, every time.

Scalar Test Results (BCS 1.0): Rivers–Vuong T, $N(0,1)$

	Others' Cost	Demographics	BLP Inst.	Dmd. Opt. Inst.
Own Profit Max vs.	Panel 1: $A(\mathbf{z}_t) = \mathbf{z}_t$, linear $h_s(\cdot)$			
Single Product	1.6489	1.1116	0.9977	1.6432
Common Ownership	-3.8928	-1.1957	0.5044	-1.0329
Double Marginalization	1.4435	0.9892	-0.0427	5.5684
Common Ownership + Double Marginalization	-0.1919	0.6815	0.1404	5.4688
Perfect Competition	1.1730	0.4171	0.7364	3.9589
Monopolist	-1.4097	-1.0680	-0.4523	-1.0908
Own Profit Max vs.	Panel 2: $A(\mathbf{z}_t) = \mathbb{E}[\Delta\eta^{12}\|\mathbf{z}_t]$, linear $h_s(\cdot)$ and $g(\cdot)$			
Single Product	1.4264	0.5795	0.6662	0.7516
Common Ownership	-2.3044	-0.5105	-0.0384	-1.4297
Double Marginalization	0.8644	0.4421	-0.5311	2.9800
Common Ownership + Double Marginalization	-0.9382	-0.2389	-0.3684	0.2460
Perfect Competition	0.7164	0.6135	-0.1080	1.8776
Monopolist	-0.8577	-0.4002	-0.3868	-1.1097
Own Profit Max vs.	Panel 3: $A(\mathbf{z}_t) = \mathbb{E}[\Delta\eta^{12}\|\mathbf{z}_t]$, random forest $h_s(\cdot)$ and $g(\cdot)$			
Single Product	5.0725	5.3347	5.2702	5.5758
Common Ownership	-3.9823	-3.6775	-4.0786	-4.3977
Double Marginalization	-6.3295	-9.9033	-6.5311	-7.6302
Common Ownership + Double Marginalization	-6.6568	-6.8735	-6.6624	-7.5578
Perfect Competition	-5.5005	-7.7749	-6.4083	-7.7653
Monopolist	-3.7240	-4.4602	-3.6134	-3.8959

Example results: RTE cereal (NielsenIQ, 2007–2017), BLP demand with micro moments; the menu is single-product, own-profit, common ownership κ, and monopoly pricing. $\mathsf{T} < 0$ favors own-profit. Every entry is conditional on the estimated demand system that produced the markups.

Full Conditional-Moment Results (BCS 2.0): Own-Profit Against Each Alternative

Studentized contrast $\Delta\widehat{Q}$ of own-profit against the row model, unified-bootstrap p-value in parentheses; $Z < 0$ favors own-profit. Product+year+chain FE, servings weights, boosting, controls partialled out of the instrument basis.

own-profit vs.	`opt_iv` Z (p)	`demog` Z (p)
single-product	$+5.22\,(< .001)$	$+4.18\,(< .001)$
common ownership κ	$-3.11\,(.005)$	$-4.09\,(< .001)$
common ownership κ_{lag}	$-2.57\,(.012)$	$-3.96\,(< .001)$
common ownership κ_{ma}	$-2.46\,(.024)$	$-4.28\,(< .001)$
monopoly (big 4)	$-3.82\,(< .001)$	$-4.82\,(< .001)$

- **Same ladder as the scalar test**: single-product beats own-profit, and own-profit beats common ownership and monopoly, under both instrument groups with power.
- The rivals' cost shifters (`opc`) have no power for any pair; the differentiation instruments (`diff`) separate single-product from own-profit ($+4.91$) and nothing else. Strength for conduct, not validity, is what limits them.
- Every $Q_m > 0$: this is "rank the misspecified models," conditional on the demand-side instruments being excluded from the true cost shock, and on the estimated demand system.

The Single-Product Puzzle

- Single-product pricing (each brand ignores its sister brands) fits the moment condition better than own-profit maximization. Robust across instruments, samples, fixed effects, weighting, learners, and both criteria; look back at the scalar-test table.
- It is not "no market power": single-product markups are $1/(\alpha(1-\sigma_j))$-sized, comfortably above cost. It says firms appear to internalize less cross-brand recapture than Bertrand with the estimated demand implies.
- Candidate explanations, none settled:
  1. Demand misspecification: the cross-price elasticities (recapture) are mismeasured, so the "correct" internalization looks wrong through the lens of this demand system. Recall the micro-data week: substitution patterns are the thing BLP gets least right.
  2. Rule-of-thumb pricing: brand managers price their own brand; the corporate parent does not reprice Cheerios when Lucky Charms moves.
  3. Another instrument or moment may reorder single vs. own-profit; nothing we have tried does.
- Methodological point: a result like this only becomes visible once the test is honest about ranking misspecified models. A specification test would have rejected everything and said nothing.

Stepping Back: What About the Mechanism?

- In RTE cereal, the evidence favors own-profit maximization over common ownership pricing, on BCS 1.0 and BCS 2.0 alike.
- In some sense this is a frictionless model
  - Managers do exactly what investment managers want
  - Investment managers seek to maximize portfolio value
- Some attempts at micro foundations are questionable
  - Anton, Ederer, Gine, Schmalz: toy model has manager choose effort to reduce mc.
  - Weaker managerial incentives: higher mc and higher p
  - Some correlation that executives at commonly owned firms have performance less correlated with profitability.
- In some sense CO is what we would see absent agency problems, so where are they?

Thanks!